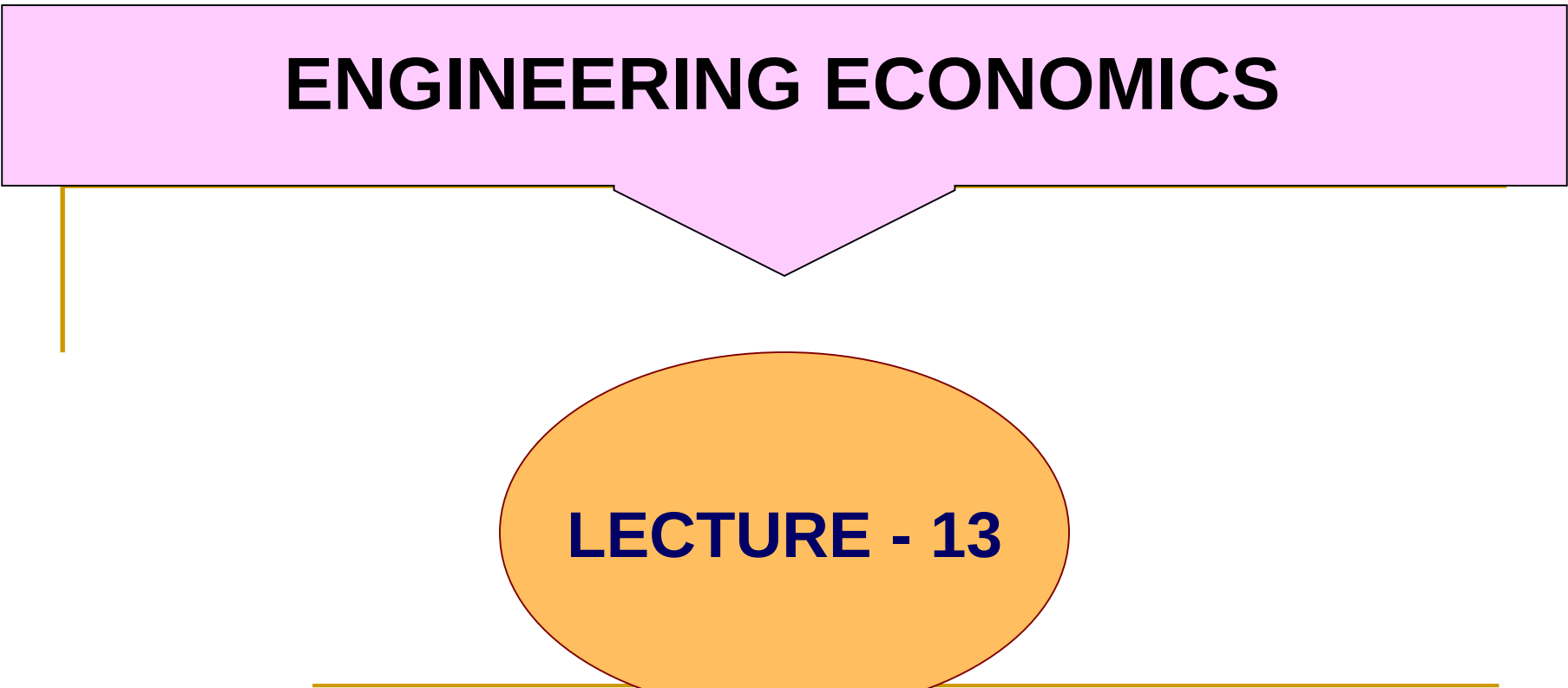


ENGINEERING ECONOMICS



LECTURE - 13

Break Even Analysis

Break-Even is the output level where total costs equal total revenue.

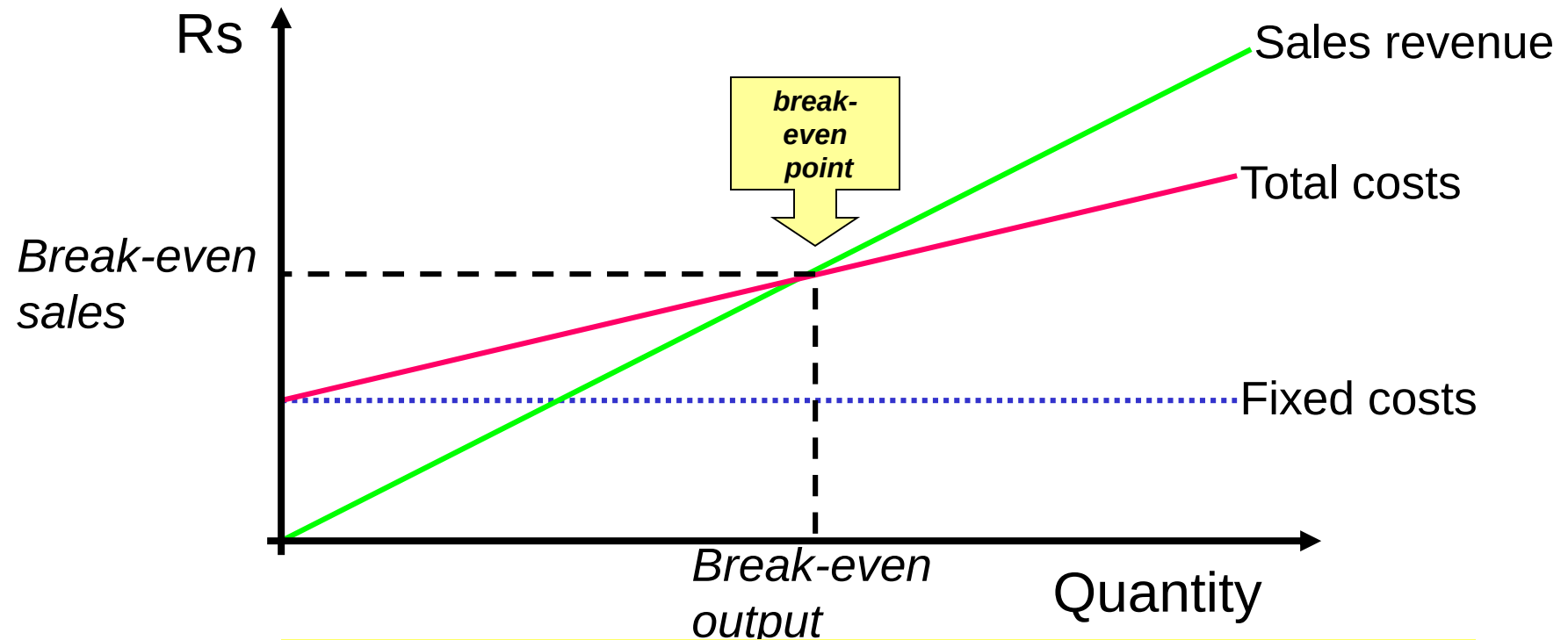
Break-even analysis is a technique used to estimate the number of units which must be produced and sold for a project to break-even.

In business, revenue or revenues is income that a company receives from its normal business activities.

Profits or net income generally imply total revenue minus total expenses in a given period.

- Break-even analysis provides a simple means of measuring profits and losses at different levels of output
- With this simple technique, a manager can calculate the effect of different marketing strategies and different pricing strategies on the business.
- The break-even point gives a business an initial target at which to aim.

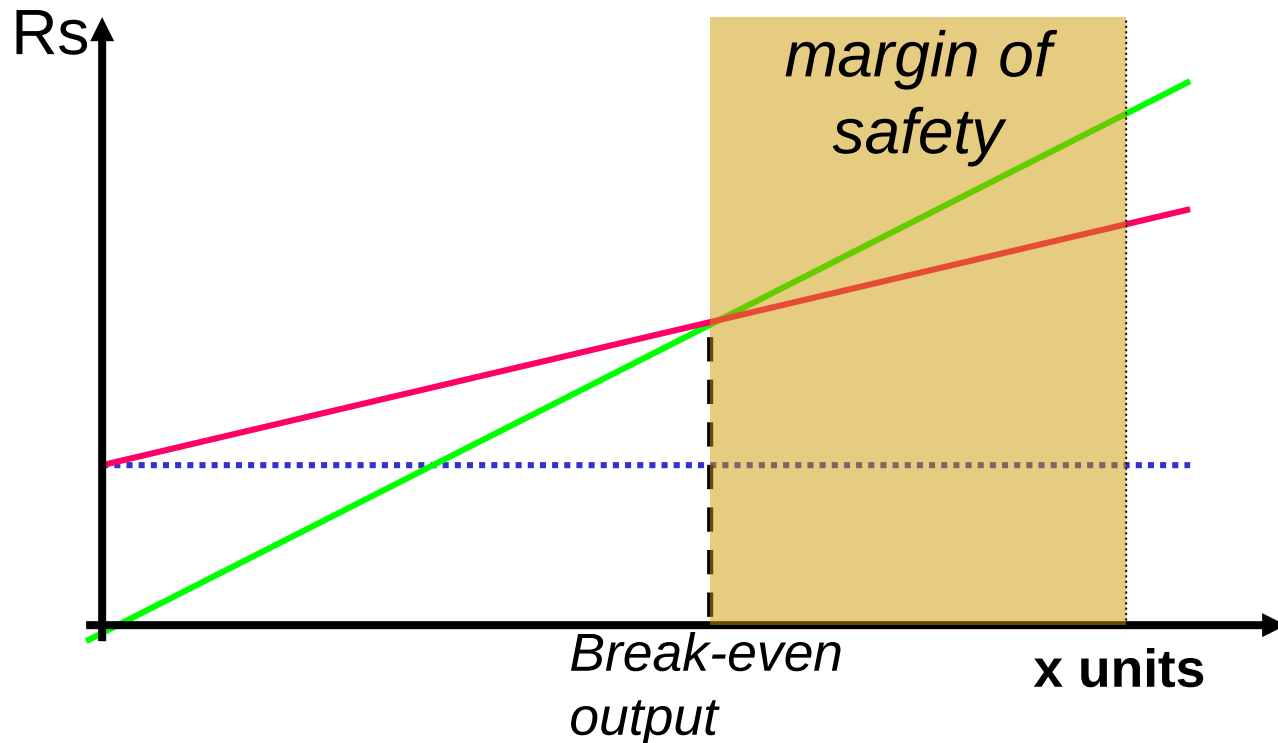
Drawing a Break-Even Chart



At the break-even point, total sales = total cost
(i.e. no profit or loss is made)

Margin of Safety

The difference between actual output and the break-even output is known as the **margin of safety**.



Break-Even Point

- The point at which total costs are covered and no profit or loss is made is called the **break-even point**
- The break-even point is where the **total revenue** and **total cost** lines intersect on the chart
- This can also be calculated using the
$$\text{BEP} = \frac{\text{fixed costs}}{\text{contribution per unit}}$$

To work out break-even we need to know various bits of information:

- The **price** you are charging
- The **variable costs** (direct costs) of each unit - these are the costs of raw materials, labour and so on that can be directly attributed to each unit.
- The **fixed costs** (or indirect costs/overheads) - these are the costs that stay the same whatever the level of output and will be things like rent, marketing costs, admin costs and so on.

Once we have this information, we can work out the break-even level of output. Let's look at an example:

Break-Even Calculation

- Assuming that a product has a selling price of £6. Variable costs are £1 per unit and fixed costs are £50,000 per year
- Calculate the number of units that a firm must sell in order to break-even
- Answer -
$$\frac{£50,000}{£5} = 10000 \text{ units}$$

Uses of Break-Even Analysis

- To calculate the minimum amount of sales required in order to be able to break even
- To see how changes in output, selling price or costs will affect profit levels
- To calculate the level of output required to reach a certain level of profit
- To allow various scenarios (what-if) to be tested out
- To aid forecasting and planning

Limitations of break-even analysis

- It's accuracy depends upon the accuracy of the data used
- Forecasting the future is difficult, especially long term
- It assumes there is a simple relationship between variable costs and sales
- Sales income does not necessarily rise in a constant relationship to sales volume
- Break-even does not specify a **time** that it will take to reach this level. This will depend on how quickly sales are generated.

Discussion